Module 2 Assignment
7nm FinFET Device and Inverter Characterization

Characterization Table

S.No	W/L (P)	W/L (N)	V_{th} (V)	I_d (μA)	P (W)	t_p (ps)	A_v	f (Hz)
1	3.00	2.00	0.372	268	2.04×10^{-15}	0.255	6.46	2.21×10^{10}
2	2.28	2.28	0.344	258	1.94×10^{-15}	0.253	6.42	2.24×10^{10}
3	2.00	2.28	0.355	242	1.81×10^{-15}	0.252	6.43	2.25×10^{10}

Unique SPICE Deck Screenshot

```

** sch_path: /home/maddy/asap_7nm_Xschem/inverter_vtc.sch
**.subckt inverter_vtc
Xpfet1 nfet_out nfet_in vdd vdd asap_7nm_pfet l=7e-009 nfin=14
Xnfet1 nfet_out nfet_in GND GND asap_7nm_nfet l=7e-009 nfin=14
V1 nfet_in GND pulse(0 0.7 20p 10p 10p 20p 500p 1)
V2 vdd GND 0.7

Vuniq in 0 DC 0.746

**** begin user architecture code


.dc v1 0 0.7 1m
*.tran 1e-12 100e-12

.control
    * First run DC
    dc v1 0 0.7 1m
    run

    * DC measurements
    meas dc v_th when nfet_out = nfet_in
    plot nfet_out nfet_in

    let gain_av = abs(deriv(nfet_out))
    meas dc max_gain max gain_av
    let gain_target = max_gain * 0.999
    meas dc vil find nfet_in when gain_av = gain_target cross=1
    meas dc voh find nfet_out when gain_av = gain_target cross=1
    meas dc vih find nfet_in when gain_av = gain_target cross=2
    meas dc vol find nfet_out when gain_av = gain_target cross=2
    let nmh = voh - vih
    let nml = vil - vol
    print v_th max_gain vil voh vih vol nmh nml

```

Figure 1: Screenshot of the modified unique SPICE deck used for the inverter simulation.

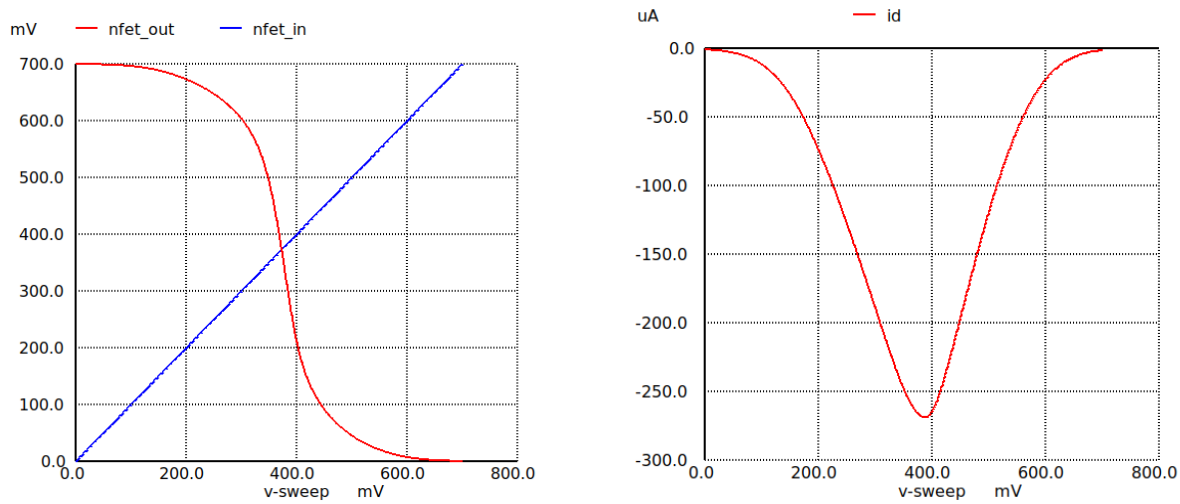

Figure 2: Voltage transfer characteristic (left) and drain current plot (right) of the 7nm FinFET inverter for PMOS/NMOS ratio of 3/2.

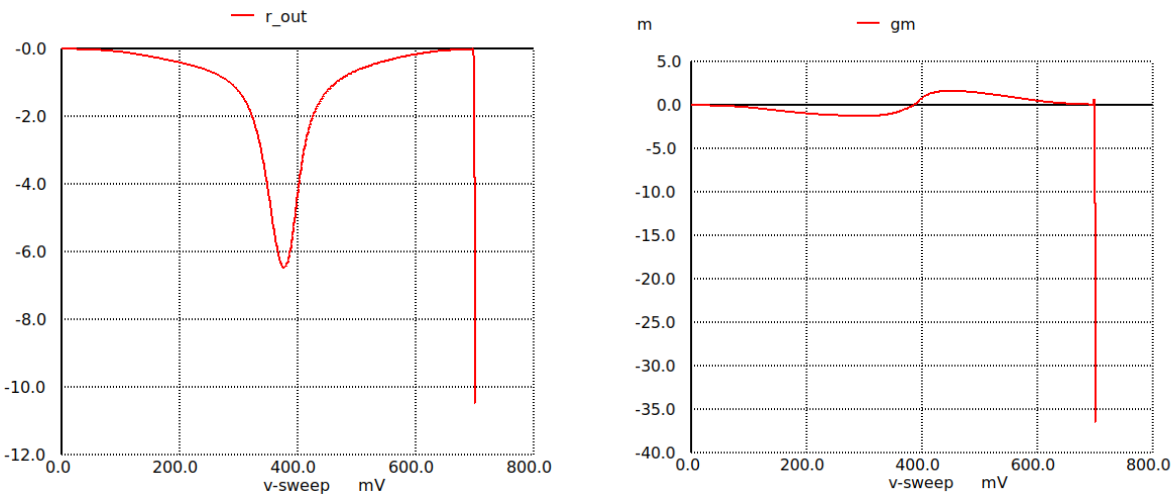

Figure 3: Output resistance (r_o) plot (left) and transconductance (g_m) plot (right) of the 7nm FinFET inverter for PMOS/NMOS ratio of 3/2.

Simulation Plots for W/L = 3/2

NGSPICE Output Screenshots for W/L = 3/2

```
v_th                  =  3.72639e-01
max_gain              =  6.46601e+00 at=  3.79000e-01
vil                   =  3.77625e-01
voh                   =  3.40722e-01
vih                   =  3.79997e-01
vol                   =  3.25383e-01
v_th = 3.726390e-01
max_gain = 6.466012e+00
vil = 3.776245e-01
voh = 3.407215e-01
vih = 3.799970e-01
vol = 3.253833e-01
nmh = -3.92755e-02
nml = 5.224120e-02
gm max                =  1.61271e-03 at=  4.50000e-01
```

```
Initial Transient Solution
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Node                                   Voltage
----                                   -------
nfet_out                              0.699765
nfet_in                                      0
vdd                                        0.7
in                                       0.746
vuniq#branch                                 0
v2#branch                           -8.07409e-07
v1#branch                            9.23848e-12


No. of Data Rows : 120
tpr                  =  2.50000e-11
tpf                  =  2.60580e-11
id_pwr               =  -2.04553e-15 from=  2.00000e-11 to=  6.00000e-11
tpr = 2.500000e-11
tpf = 2.605796e-11
tp = 2.552898e-11
id_pwr = -2.04553e-15
pwr = -1.43187e-15
power = 3.579676e-05
```

Figure 4: NGSPICE output screenshots showing the simulation results and extracted parameters for the 7nm FinFET inverter with PMOS/NMOS ratio of 3/2.